

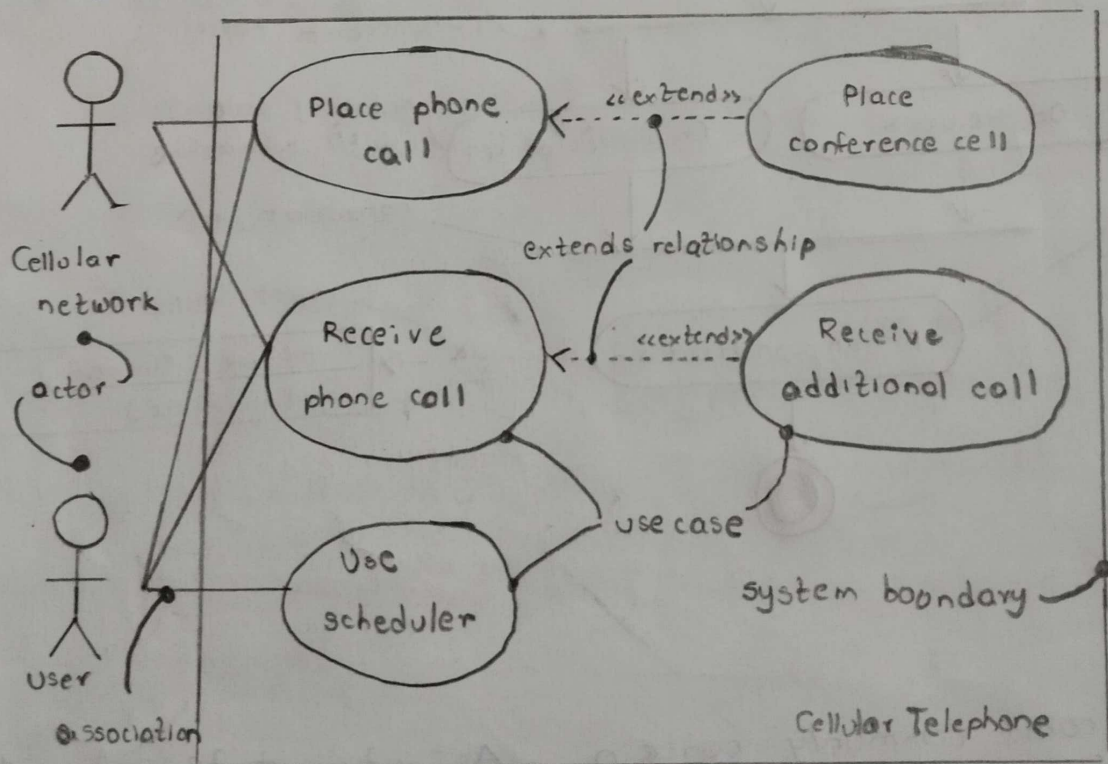
# ASSIGNMENT-IV

1. Explain a simple use case diagram with an example.

Ans: USE CASES:

A use case describes a set of sequences, in which each sequence represents interaction of things outside system (actors) with system itself (key abstractions).

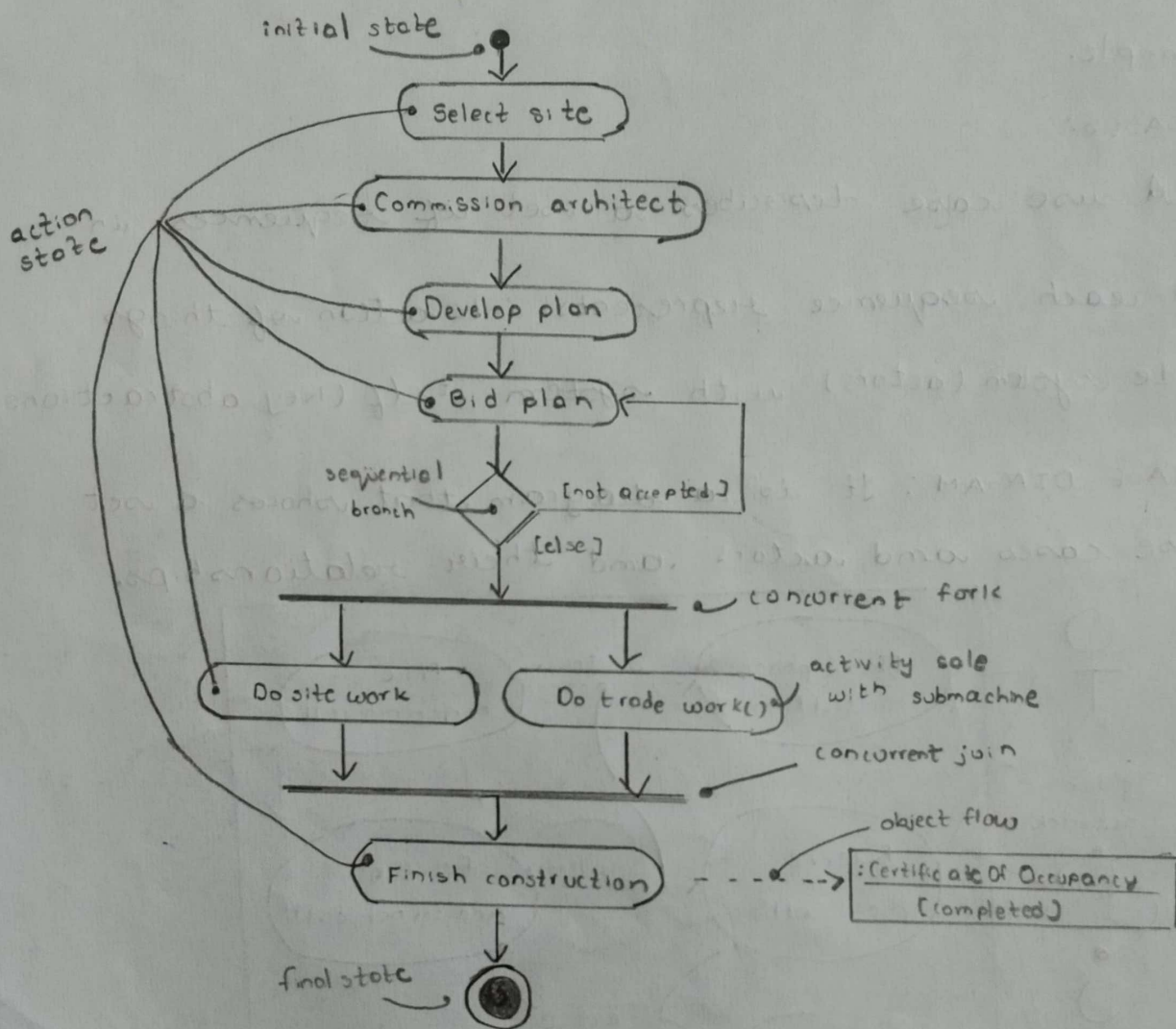
USE CASE DIAGRAM: It is a diagram that shows a set of use cases and actors and their relationships.



\* A use case diagram is just a special kind of diagram and shares the common properties as do all other diagrams.

2. Explain an activity diagram with an example.

Ans Activity Diagram: Activity diagram is used to model dynamic aspects.



Contents:

Activity diagram commonly contain . Activity state and action states

- Transitions

- Objects

3. List the key features required for a Vacation Tracking system (VTS).

Ans. Here's a breakdown of primary features.

#### 1. User Management

Classes: User, Employee, Manager, HR admin

Features: Employee, Registration/Login, Role based Access.

#### 2. Vacation Request Management

Classes: Vacation Request, Leave type

Features: Create, Submit and modify vacation request

#### 3. Approval Workflow

Classes: Approval process, notification

Feature: Manager approval/rejection system

#### 4. Leave Balance Tracking

Classes: Leave Balance, Leave History

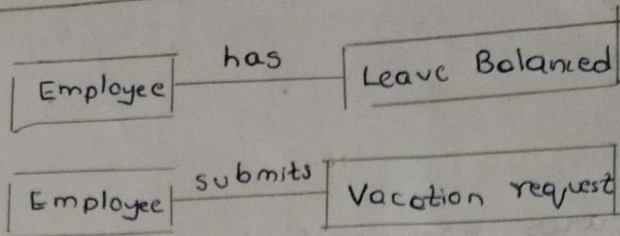
Features: Track accrued leave and used leave.

#### 5. Calendar Integration

Classes: Calendar, Holiday

Features: Display vacations on a shared calendar

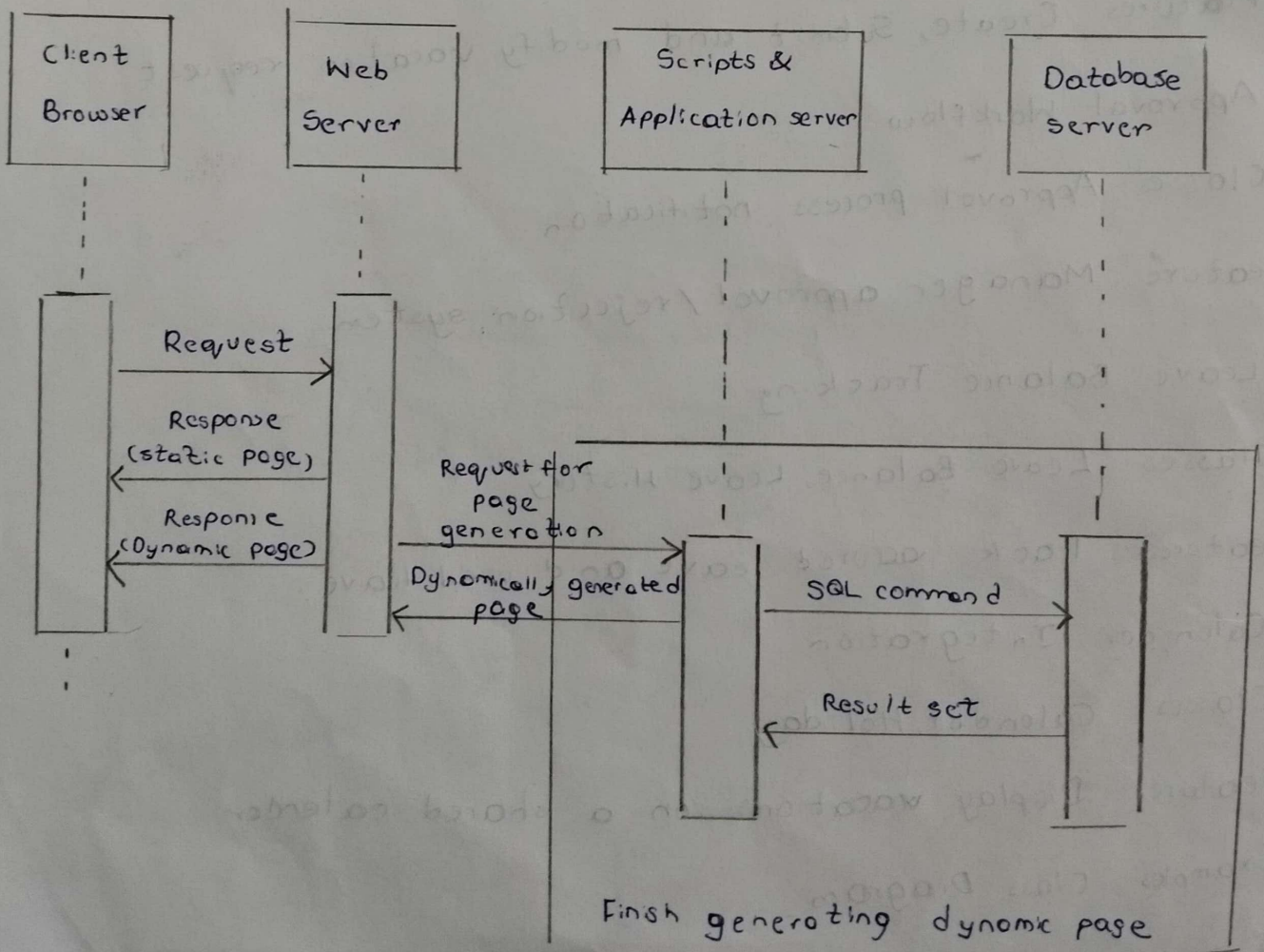
Examples Class Diagram.



4. Explain the top-level use case model for Vacation Tracking System (VTS).

Ans. A Top-level usecase model for a vacation tracking system (VTS) would primarily includes three key factors: "Employee", "Manager" and "HR administration".

Request GET & post

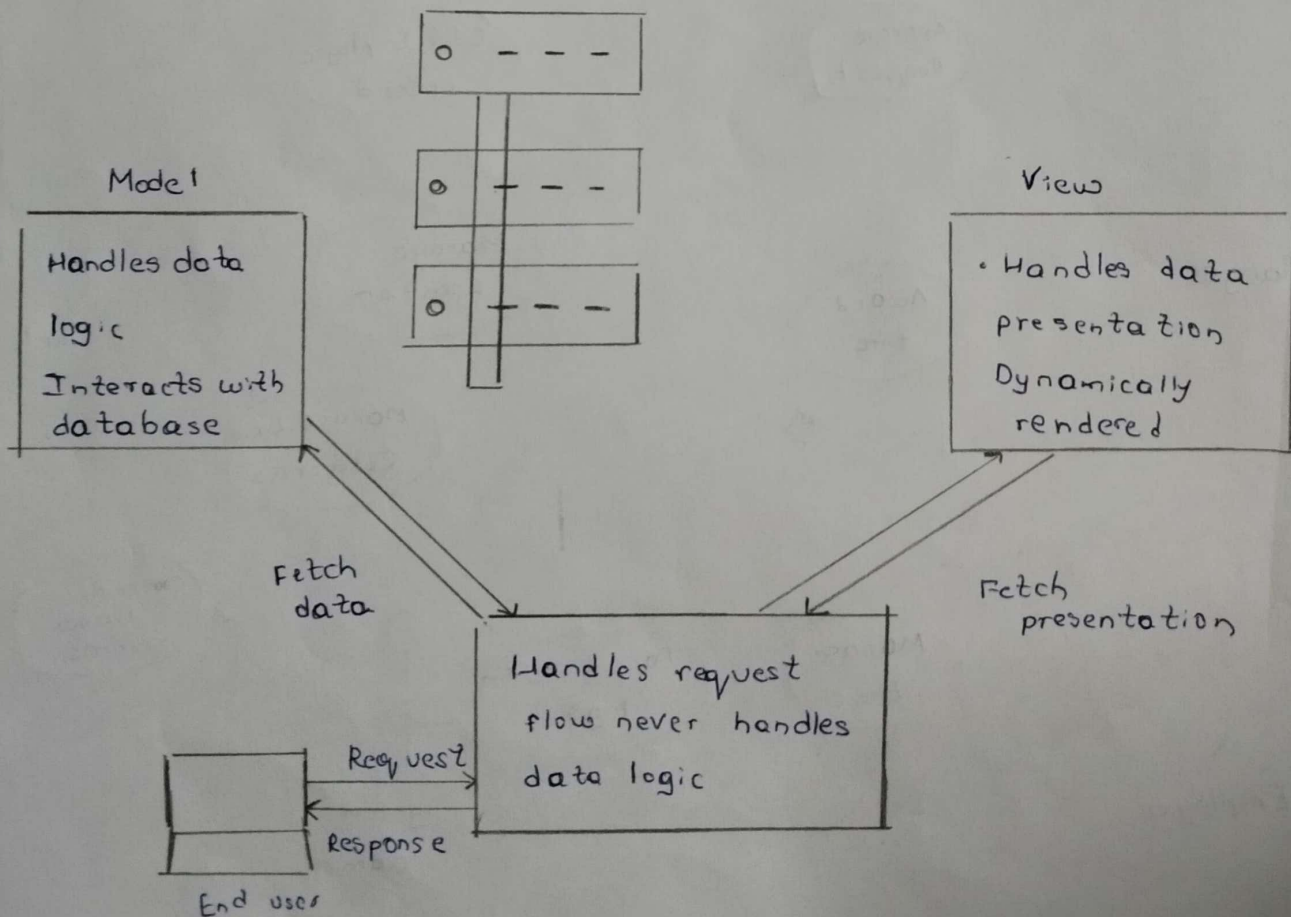


Draw the basic sequence diagram of an HTTP GET Request for a web application.

Ans. A sequence diagram for a web application of an HTTP GET Request

1) User Load Application

- [User] --- "Request Resource()" ----> [Web Browser]
- [Web Browser] --- "Request Access For User()" ----> [Web Server]
- [Web Server] --- "Check User EXISTS()" ----> [Content Database]
- [Content Database] --- "Exists Return Message" ----> [Web Server]



6. Explain the role of Controllers in MVC web architecture.

Ans. A controller is responsible for controlling the way that a user interacts with an MVC application. A controller contains the flow control logic for an ASP.NET MVC application. A controller determines what response to send back to a user when a user makes a browser request.

The primary functions of the requesting, reviewing and managing vacation time within the system.

